

Solution Requirements

Date	02 July 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users can access the HDI prediction system through the web application without requiring login.	Low
2	Authorization Levels	All users have permission to enter development indicators and view prediction results.	Low
3	External Interfaces	The system uses the trained ML model (<code>HDI.pkl</code>) and reads the HDI dataset for prediction.	High
4	Transactions Processing	Users enter Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI per Capita, and the system predicts the HDI score instantly.	High
5	Reporting	Display the predicted HDI score, HDI category, and graphical visualizations for analysis.	Medium
6	Business Rules	Validate input values and classify HDI as Low, Medium, High, or Very High based on the predicted score.	High
7	Compliance to Laws or Regulations	Use only publicly available HDI data and do not store any personal user information.	Medium
8	Other	Provide a responsive web interface with Home, Prediction, and About pages.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Generate HDI predictions quickly after user input.	Prediction response time < 2 seconds
2	Scalability	Support multiple users making predictions simultaneously.	Supports 50 concurrent requests
3	Security & Data Privacy	No personal data is stored; only prediction inputs are processed in memory.	No user data stored
4	Reliability & Availability	The application should run continuously without failures while the Flask server is active.	99% uptime during usage
5	Usability & Accessibility	Provide a simple, responsive interface that works on desktop and mobile devices.	Responsive layout, easy-to-use forms
6	Other (Maintainability)	Separate Training and Flask modules, with clear documentation for retraining and deployment.	Well-documented project structure with README